



RAMCloud:

The Future of Storage Systems

In RAMCloud, data is stored in the DRAM of thousands of computers in a data centre. RAMCloud offers quick and reliable recovery even though terabytes of data may be stored in the system.

Today, the amount of data generated on the Internet is enormous. An application like Facebook needs to deal with petabytes or terabytes of data without compromising its performance. RAMCloud is a next generation storage system that can deliver high performance with the help of just commodity hardware, even at current levels of storage complexity. It stores data entirely in DRAM (main memory), and the disk takes the role of backup or archival storage. Since the data always resides in the main memory, it can avoid the access latency that is usually incurred in a disk-based storage system and, hence, provides high throughput, which is the key to better performance.

The current scenario

For the past four decades we have seen rapid growth in computer hardware technologies, which has helped improve the efficiency of storage systems. The processor, memory and disk play an important role in the performance of a storage system. Currently, an imbalance in the performance of any one of these components can impact the whole system.

Though there has been tremendous improvement in the performance of both memory and processor, the disk has not been able to keep pace. Disk capacity has increased by more than a thousand times, but the transfer rate for large blocks has improved only fifty-fold, while seek time and rotational latency have only improved two-fold.

Large applications like Facebook and Amazon require multiple access points to storage servers to generate a single page. Due to the high access rate the performance of these applications is reduced. Applications use cache to overcome disk latency, but cache must have an exceptionally high hit rate to provide significant performance improvement. Even a 1 per cent cache miss can severely affect the system's performance, which is not acceptable for some applications.

Flash memory is another storage system that offers latency lower than disk. But Flash devices are I/O devices, so apart from the access latency, they have additional latencies of device drivers and interrupt handlers. These shortcomings of current storage systems demand a new improved storage approach.